



<https://youtu.be/cNEVDsU9xx0>

GNUstep

Architectural Enhancement

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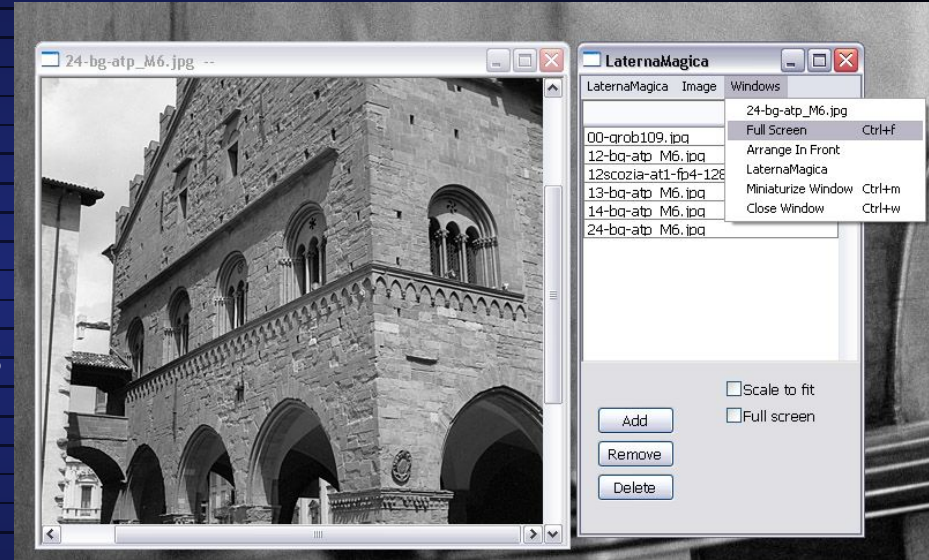
Abstract

The report outlines a proposed enhancement to the GNUstep framework, with detailed explanation on its benefits, the necessary architectural modifications, and how it interacts with existing features. It also includes planned testing, risk analysis, and an SEI SAAM architectural analysis that identifies key stakeholders and non-functional requirements. Two implementation approaches are presented with their respective impacts, and the report highlights the best option, provides use cases with sequence diagrams, and concludes with lessons learned and final conclusions.



Introduction

Developing a desktop application demands appealing visuals, yet GNUstep's current theming—rooted in 1990s NeXTstep and outdated Windows styles—feels outmoded. Since theming affects crucial UI elements like buttons, menus, fonts, and check-boxes, we propose enhancing GNUstep's theming packages to align with modern design standards.



Proposed Enhancement

Introducing a CSS-like theming system, high-DPI scaling, and system UI integration while enabling customizable widget styles, adaptive padding, and modern UI components for an enhanced user experience. Our proposed enhancement is to update and modernize the theming options for desktop applications made with the GNUMstep framework.

Value and Benefit of Feature

Modernizing GNUMstep's theming options would allow developers to create desktop applications that reflect modern aesthetics, eliminating design restrictions and making programs more accessible. This update benefits both experienced GNUMstep users and new developers by enabling the creation of visually appealing, contemporary applications.

Changes to Architecture

The enhancement will mainly affect the `libs-gui` library, which handles the visual theming of GNUstep applications. Initially, the update involves adding new visual assets and updating files in the existing theming package. If further theme customization features are introduced, both `libs-gui` and `libs-back` will require modifications—`libs-gui` to ensure proper rendering on the front end, and `libs-back` to support the new functionality and manage additional assets.

Interactions with Other Features

The proposed enhancements will significantly impact GNUstep's core libraries. They involve extending classes and functions in GNUstep-gui and GNUstep-back to apply CSS-like style rules to UI elements, updating layout management for dynamic theme adjustments, and overhauling the theming system. Additionally, rendering and scaling modifications in GNUstep-back will ensure proper font and UI alignment across various resolutions, while maintaining backward compatibility with non-CSS styled applications.

Planned Tests

The tests will evaluate the enhancements' impact on compatibility, performance, and user experience while ensuring system stability. The CSS-like theming system must integrate with existing UI components and maintain backward compatibility with older themes. High DPI scaling will be tested for responsiveness and visual clarity across various resolutions, and customizable widget styles will be verified to ensure layouts and accessibility remain intact across different screen sizes and DPI settings.

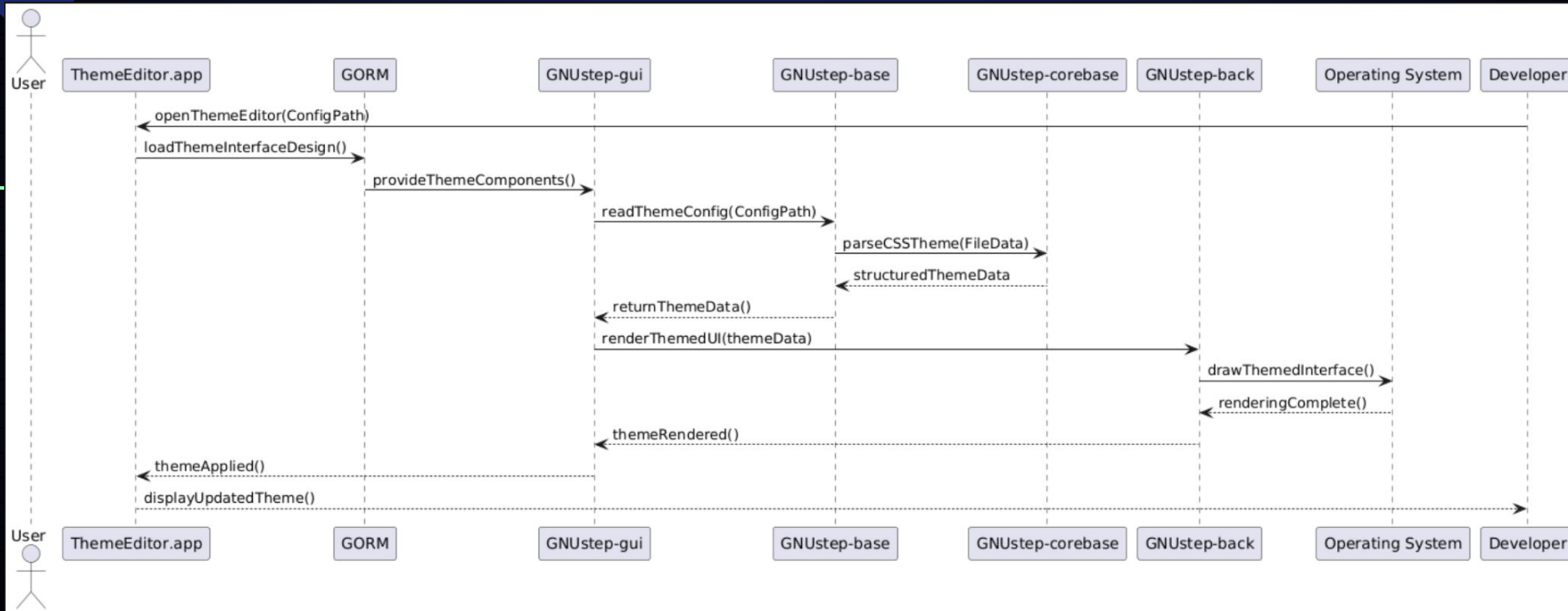
Risk Analysis

The enhancement poses significant risks to GNUstep, including potential security vulnerabilities from major subsystem changes, increased maintainability challenges, and the possibility of long-lasting bugs due to a slow, volunteer-based development cycle. Its large scope may overwhelm the small developer team, risking incomplete implementation and further alienating users.

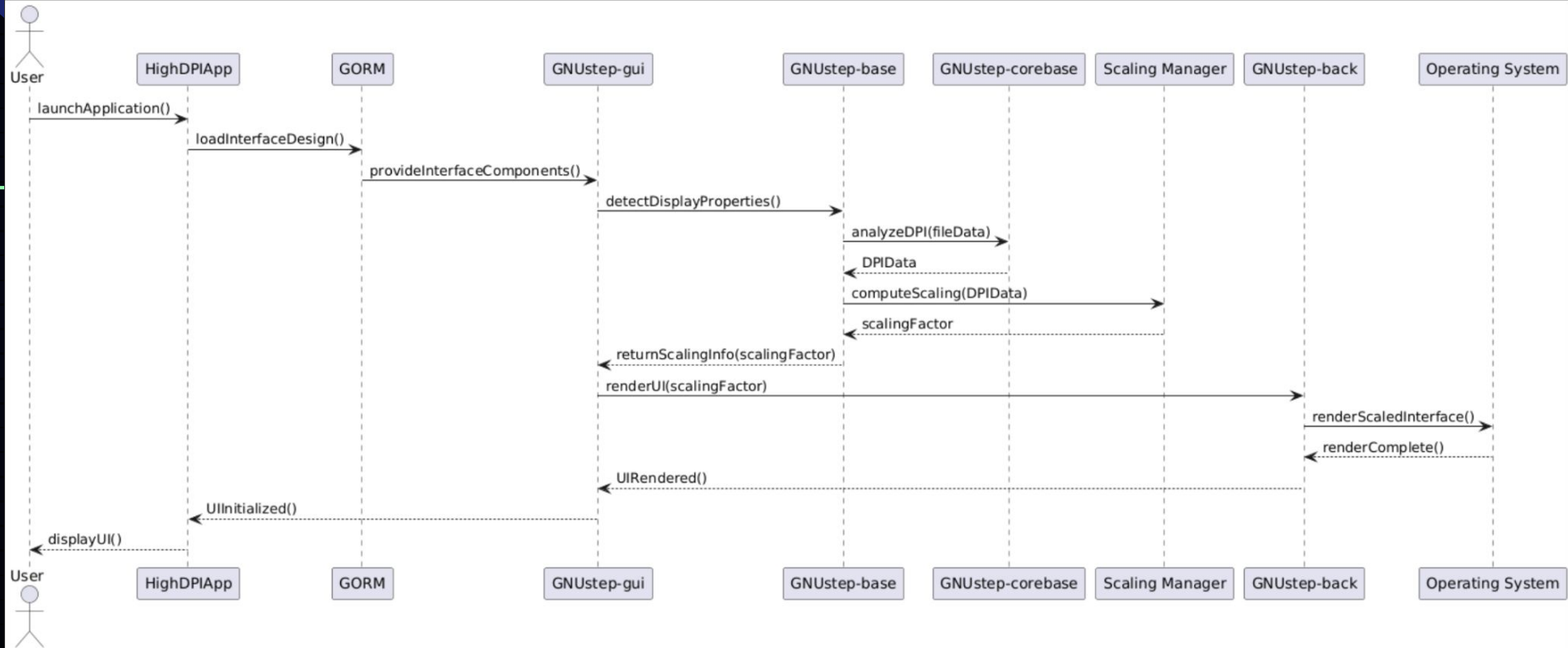
Software Architecture Analysis Method

Improved UI development tools mainly benefit developers by enabling modern, visually appealing applications. Two enhancement methods are discussed: CSS-based theming, which offers high customizability by letting developers design their own themes, and system UI integration, which automatically adapts to OS themes but limits customization. Ultimately, CSS-like theming is favored for its flexibility despite its higher complexity.

USE CASES: Theme Editor



USE CASES: High DPI



Lessons Learned

Our exploration of GNUStep enhancements taught us the value of deep system understanding and prior research to identify useful improvements. We realized that recognizing the system's limitations is important for proposing the system modernization and creating a detailed implementation plan. Additionally, modifying a large codebase is intricate where effective enhancements must consider complex component interactions, potential changes to non-functional requirements, stakeholder impacts, and multiple implementation alternatives.



Data Dictionary

**CSS-like
theming**



**Backwards
Compatibility**

**System UI
integration**

High-DPI

SAAM

Widgets & scaling

**Adaptive
Padding**



Rendering

**Non-Functional
requirements**



CONCLUSION

In summary, the GNUSstep enhancement modernizes outdated theming by introducing a CSS-like system, high-DPI scaling, and improved UI integration. This upgrade allows developers to create applications with modern visuals and better accessibility across various resolutions, while ensuring smooth integration and backwards compatibility through changes to libs-gui and libs-back. Although risks like security vulnerabilities, maintainability challenges, and a limited volunteer workforce exist, the benefits of a modern, flexible, and developer-friendly framework are clear. Furthermore, our exploration of GNUSstep enhancements revealed that deep system understanding and careful planning are essential when modernizing a large codebase. Ultimately, this enhancement addresses current limitations and lays a strong foundation for future improvements.

